

(TR-104) PYTHON WITH AI/ML

Training Day 37 Report

Date: 3 March 2026

Topic Covered: LangChain Integration, Vector Database Storage & Local LLM Setup for RAG System

The thirty-seventh day of training focused on extending the RAG-based AI Assistant project by integrating LangChain for document handling, implementing vector database storage using ChromaDB, and setting up a local Large Language Model using Ollama and Mistral.

- Loaded TXT, PDF, and HTML documents using LangChain document loaders.
- Compared manual document loading with LangChain-based document processing workflows.
- Combined extracted documents into a unified document collection for processing.
- Implemented document chunking using RecursiveCharacterTextSplitter for efficient retrieval.
- Generated semantic embeddings using SentenceTransformer MiniLM model.
- Stored document embeddings and metadata inside ChromaDB vector database.
- Implemented similarity search for retrieving top-k most relevant document chunks.
- Understood how semantic retrieval works inside RAG systems.
- Installed Ollama for running local Large Language Models.
- Configured and used the Mistral model locally through Ollama.
- Learned how local LLMs can be integrated with retrieval systems for private AI workflows.
- Strengthened understanding of LangChain pipelines, vector databases, embedding storage, and retrieval-based AI systems.
- Explored the workflow of combining retrieval systems with local LLM inference for contextual response generation.

- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/projects>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/understanding-rag-how-retrieval-enhances-ai-generation-5a0c79830cbc>